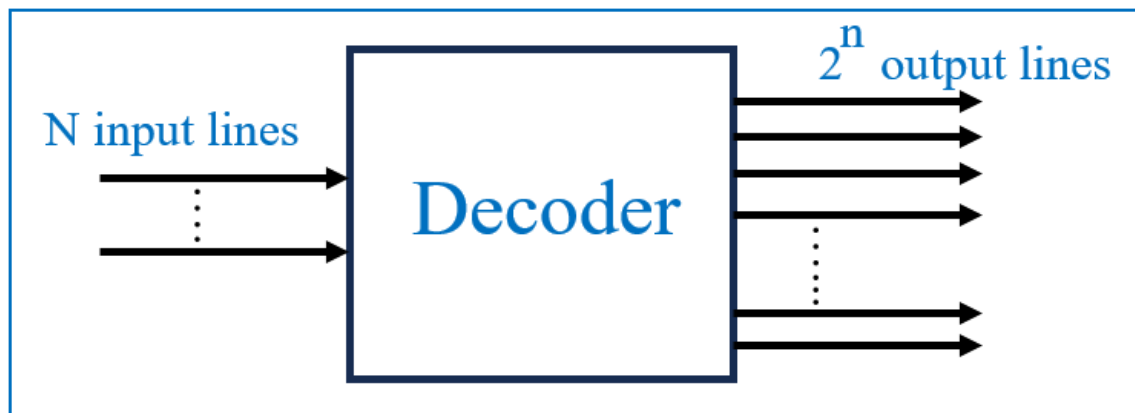


Day 14

2x4 and 3x8 Decoders in Verilog

Circuit Diagram :



Verilog Code

A) 2:8 Decoder code

```
1  module decoder2x4 (sel,Y);
2      input [1:0] sel;
3      output reg [3:0] Y;
4      always @(*) begin
5          case (sel)
6              2'b00: Y = 4'b0001;
7              2'b01: Y = 4'b0010;
8              2'b10: Y = 4'b0100;
9              2'b11: Y = 4'b1000;
10         endcase
11     end
12 endmodule
13
```

B) 3:8 Decoder code

```
1 module decoder3x8(sel,Y);
2     input [2:0] sel;
3     output reg [7:0] Y;
4     always @(*) begin
5         case (sel)
6             3'b000: Y = 8'b00000001;
7             3'b001: Y = 8'b00000010;
8             3'b010: Y = 8'b00000100;
9             3'b011: Y = 8'b00001000;
10            3'b100: Y = 8'b00010000;
11            3'b101: Y = 8'b00100000;
12            3'b110: Y = 8'b01000000;
13            3'b111: Y = 8'b10000000;
14        endcase
15    end
16 endmodule
```

Testbench Code

A) 2:4 Decoder Tb

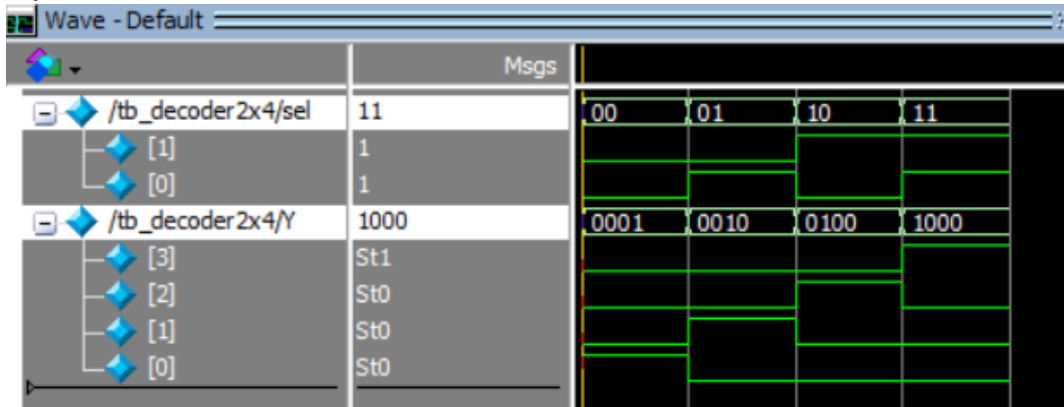
```
1 module tb_decoder2x4;
2     reg [1:0] sel;
3     wire [3:0] Y;
4     decoder2x4 dut (sel,Y);
5     initial begin
6         $monitor("Time=%0t sel= %b Y=%b", $time, sel, Y);
7         sel = 2'b00; #10;
8         sel = 2'b01; #10;
9         sel = 2'b10; #10;
10        sel = 2'b11; #10;
11        $finish;
12    end
13 endmodule
```

B) 3:8 Decoder Tb

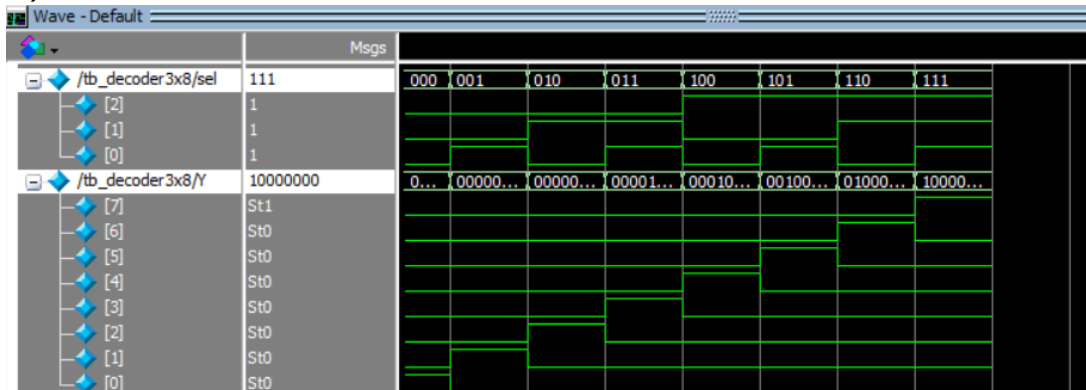
```
1 module tb_decoder3x8;
2     reg [2:0] sel;
3     wire [7:0] Y;
4     decoder3x8 dut(sel,Y);
5     initial begin
6         $monitor("Time=%0t sel=%b Y=%b", $time,sel,Y);
7         sel = 3'b000; #10;
8         sel = 3'b001; #10;
9         sel = 3'b010; #10;
10        sel = 3'b011; #10;
11        sel = 3'b100; #10;
12        sel = 3'b101; #10;
13        sel = 3'b110; #10;
14        sel = 3'b111; #10;
15        $finish;
16    end
17 endmodule
```

Waveform

A)

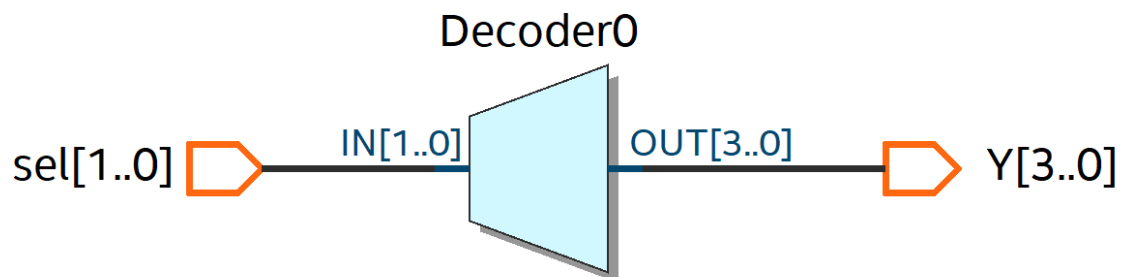


B)

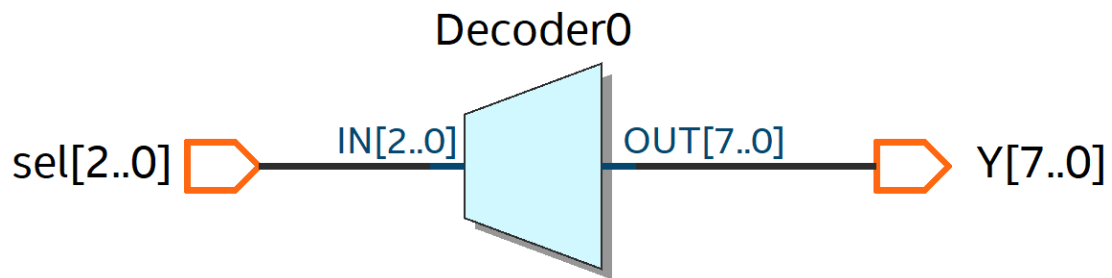


Schematic

A)



B)



EDA Tools Used

- IntelQuartusPrime
- ModelSim

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